

On the KKT Method

Agni Datta*

University of Edinburgh, UK
agnidatta.org@gmail.com

Abstract. We study the Karush–Kuhn–Tucker method as the basic language of constrained nonlinear optimization. We start with Lagrange multipliers for equality constraints and then explain how inequality constraints introduce signs, complementary slackness, tangent cones, and normal cones. We prove the KKT necessary conditions under standard constraint qualifications, prove the convex sufficiency theorem, and relate KKT points to saddle points and strong duality. We then discuss second-order conditions, strict complementarity, sensitivity, complementarity problems, sequential quadratic programming, active-set methods, and interior-point methods. The point of the note is to show that KKT is not a collection of equations to memorize; it is the common certificate behind duality, stationarity, feasibility, and local optimality [Kar39, KT51, Joh48, Roc70, Ber99, NW06, BV04].

1 Introduction

The classical method of Lagrange multipliers explains equality-constrained optima by saying that the objective gradient must lie in the span of the constraint gradients. Karush and later Kuhn and Tucker extended this picture to inequality constraints, where a constraint may be active or inactive, and where an active inequality can push only in one signed direction [Kar39, KT51]. Fritz John’s theorem placed these conditions in a more general necessary-condition framework that does not require the same regularity assumptions but may introduce a degenerate multiplier on the objective [Joh48]. The modern KKT theorem is the regular case in which the multiplier on the objective can be normalized to one.

This historical distinction matters. The KKT equations are often written as a checklist, but the theorem is a separation statement about when first-order local optimality can be separated from feasible first-order motion. Constraint qualifications such as linear independence, Mangasarian–Fromovitz, and Slater’s condition are precisely the hypotheses that make this separation faithful [MF67, Sla50, Ber99, NW06]. Without such hypotheses, local optima may exist without ordinary KKT multipliers, and the right necessary conditions are Fritz John conditions or more general normal-cone inclusions.

The same equations later became central in convex analysis and numerical optimization. Rockafellar’s convex analysis made normal cones, subgradients, and saddle points the natural language of constrained convex optimization [Roc70]. Arrow, Hurwicz, and Uzawa developed

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primal-dual multiplier methods [AHU58]. Sequential quadratic programming, augmented Lagrangian methods, active-set methods, and interior-point methods all use KKT systems as their local model or their target equations [FM68, Hes69, Pow69, Han77, Pow78, NW06]. Thus, KKT is simultaneously a theorem, a certificate, and an algorithmic template.

2 The Optimization Problem

We consider the nonlinear program

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) \quad \text{subject to} \quad g_i(\mathbf{x}) \leq 0 \text{ for } i \in [m], \quad h_j(\mathbf{x}) = 0 \text{ for } j \in [p], \quad (2.1)$$

where the functions f, g_i, h_j are continuously differentiable unless a stronger smoothness assumption is explicitly stated. The feasible set is

$$\mathcal{F} = \{\mathbf{x} \in \mathbb{R}^n : g_i(\mathbf{x}) \leq 0 \text{ for all } i, \quad h_j(\mathbf{x}) = 0 \text{ for all } j\}. \quad (2.2)$$

At a feasible point $\mathbf{x}$, the active inequality set is

$$\mathcal{A}(\mathbf{x}) = \{i \in [m] : g_i(\mathbf{x}) = 0\}. \quad (2.3)$$

The active set is the bridge between inequalities and equalities. An inactive constraint has slack and cannot affect first-order local motion. An active inequality behaves like an equality for feasible first-order directions, except that its multiplier must have a sign because the constraint can block motion only toward infeasibility.

3 Equality Constraints and Lagrange Multipliers

We begin with equality constraints because they isolate the geometric idea without complementary slackness. Consider

$$\min f(\mathbf{x}) \quad \text{subject to} \quad h_j(\mathbf{x}) = 0 \text{ for } j \in [p]. \quad (3.1)$$

Let $\mathbf{h} = (h_1, \dots, h_p)$. If the gradients $\nabla h_1(\mathbf{x}^*), \dots, \nabla h_p(\mathbf{x}^*)$ are linearly independent, then the feasible first-order directions at $\mathbf{x}^*$ are the vectors $\mathbf{d}$ satisfying

$$\nabla h_j(\mathbf{x}^*)^\top \mathbf{d} = 0 \quad \text{for every } j \in [p]. \quad (3.2)$$

At a local minimum, the objective cannot decrease in any feasible first-order direction, and so $\nabla f(\mathbf{x}^*)^\top \mathbf{d} = 0$ for every direction satisfying Equation (3.2). Linear algebra then forces $\nabla f(\mathbf{x}^*)$ to lie in the span of the constraint gradients.

Theorem 3.1 (Equality-constrained multiplier rule). Let $\mathbf{x}^*$ be a local minimizer of Equation (3.1). Suppose the gradients $\nabla h_1(\mathbf{x}^*), \dots, \nabla h_p(\mathbf{x}^*)$ are linearly independent. Then there exists $\nu^* \in \mathbb{R}^p$ such that

$$\nabla f(\mathbf{x}^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(\mathbf{x}^*) = 0. \quad (3.3)$$

┘

Proof. The proof is a projection argument. The constraint gradients span the normal space to the

equality manifold, while feasible first-order directions form the tangent space. At a local minimum, the objective gradient is orthogonal to the tangent space; hence it belongs to the normal space.

Let

$$T = \{\mathbf{d} \in \mathbb{R}^n : \nabla h_j(\mathbf{x}^*)^\top \mathbf{d} = 0 \text{ for all } j \in [p]\}.$$

Under the rank assumption, the implicit function theorem identifies T with the tangent space of the equality-feasible set at $\mathbf{x}^*$. If there were a vector $\mathbf{d} \in T$ with $\nabla f(\mathbf{x}^*)^\top \mathbf{d} < 0$, then a differentiable feasible curve through $\mathbf{x}^*$ with derivative $\mathbf{d}$ would decrease f for small positive time, contradicting local optimality. Applying the same reasoning to $-\mathbf{d}$ gives $\nabla f(\mathbf{x}^*)^\top \mathbf{d} = 0$ for all $\mathbf{d} \in T$.

Thus, $\nabla f(\mathbf{x}^*)$ belongs to $T^\perp$. By definition, T is the nullspace of the matrix whose rows are $\nabla h_j(\mathbf{x}^*)^\top$, and the fundamental theorem of linear algebra gives

$$T^\perp = \text{span}\{\nabla h_1(\mathbf{x}^*), \dots, \nabla h_p(\mathbf{x}^*)\}.$$

Hence, there are scalars α_j such that $\nabla f(\mathbf{x}^*) = \sum_j \alpha_j \nabla h_j(\mathbf{x}^*)$. Setting $\nu_j^* = -\alpha_j$ gives [Equation \(3.3\)](#). $\square$

This theorem already contains the template for KKT. The only missing features are the active-set logic and the sign restriction on inequality multipliers.

4 Tangent and Normal Cones

The cleanest way to write optimality conditions is to separate feasible directions from normals. At a feasible point $\mathbf{x}$, the tangent cone records first-order directions that can be approximated by feasible displacements. For the smooth problem in [Equation \(2.1\)](#), the linearized cone is

$$\mathcal{L}_{\mathcal{F}}(\mathbf{x}) = \{\mathbf{d} : \nabla g_i(\mathbf{x})^\top \mathbf{d} \leq 0 \text{ for } i \in \mathcal{A}(\mathbf{x}), \nabla h_j(\mathbf{x})^\top \mathbf{d} = 0 \text{ for } j \in [p]\}. \quad (4.1)$$

The polar cone of a cone K is

$$K^\circ = \{\mathbf{v} : \mathbf{v}^\top \mathbf{d} \leq 0 \text{ for all } \mathbf{d} \in K\}. \quad (4.2)$$

For a minimizer, the direction $-\nabla f(\mathbf{x}^*)$ cannot be a feasible descent direction, and the gradient must therefore lie in the polar of the feasible tangent cone.

The polar of the linearized cone in [Equation \(4.1\)](#) has the form

$$\mathcal{L}_{\mathcal{F}}(\mathbf{x})^\circ = \left\{ \sum_{i \in \mathcal{A}(\mathbf{x})} \lambda_i \nabla g_i(\mathbf{x}) + \sum_{j=1}^p \nu_j \nabla h_j(\mathbf{x}) : \lambda_i \geq 0 \right\}, \quad (4.3)$$

provided the usual finite-dimensional alternatives apply without degeneracy. Constraint qualifications are precisely what lets us replace the true tangent cone by the linearized cone and then use [Equation \(4.3\)](#).

5 Constraint Qualifications

Constraint qualifications are not technical decoration. They identify when first-order linearization describes feasible first-order geometry accurately enough for multipliers to exist. Three conditions

occur constantly in the KKT theory.

Definition 5.1 (Linear independence constraint qualification). The linear independence constraint qualification, abbreviated LICQ, holds at a feasible point $\mathbf{x}$ if the gradients

$$\{\nabla g_i(\mathbf{x}) : i \in \mathcal{A}(\mathbf{x})\} \cup \{\nabla h_j(\mathbf{x}) : j \in [p]\}$$

are linearly independent. ┘

Definition 5.2 (Mangasarian–Fromovitz constraint qualification). The Mangasarian–Fromovitz constraint qualification, abbreviated MFCQ, holds at a feasible point $\mathbf{x}$ if the equality gradients $\nabla h_j(\mathbf{x})$ are linearly independent and there exists a direction $\mathbf{d}$ such that

$$\nabla h_j(\mathbf{x})^\top \mathbf{d} = 0 \text{ for all } j \quad \text{and} \quad \nabla g_i(\mathbf{x})^\top \mathbf{d} < 0 \text{ for all } i \in \mathcal{A}(\mathbf{x}).$$

Definition 5.3 (Slater condition). For a convex problem with affine equalities, Slater’s condition holds if there is a point $\mathbf{x}$ satisfying the equalities and satisfying every nonlinear inequality strictly. ┘

LICQ is strong and gives uniqueness of multipliers. MFCQ is weaker and still gives existence and boundedness properties for multipliers [MF67, NW06]. Slater’s condition is the convex-analytic condition that yields strong duality and KKT sufficiency in convex optimization [Sla50, Roc70, BV04].

6 The KKT Conditions

Define the Lagrangian

$$\mathcal{L}(\mathbf{x}, \lambda, \nu) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^p \nu_j h_j(\mathbf{x}). \quad (6.1)$$

The KKT conditions for Equation (2.1) at a candidate point $\mathbf{x}^*$ ask for multipliers $\lambda^* \in \mathbb{R}^m$ and $\nu^* \in \mathbb{R}^p$ satisfying

$$\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \lambda^*, \nu^*) = 0, \quad (6.2)$$

$$g_i(\mathbf{x}^*) \leq 0 \quad \text{for all } i, \quad h_j(\mathbf{x}^*) = 0 \quad \text{for all } j, \quad (6.3)$$

$$\lambda_i^* \geq 0 \quad \text{for all } i, \quad (6.4)$$

$$\lambda_i^* g_i(\mathbf{x}^*) = 0 \quad \text{for all } i. \quad (6.5)$$

Stationarity says that the objective gradient is balanced by the active constraint gradients. Primal feasibility says that the point belongs to the original feasible set. Dual feasibility gives the sign of inequality forces. Complementary slackness says that a constraint with slack cannot exert force.

Theorem 6.1 (KKT necessity). Let $\mathbf{x}^*$ be a local minimizer of Equation (2.1). Suppose that a constraint qualification such as LICQ or MFCQ holds at $\mathbf{x}^*$. Then there exist multipliers $\lambda^* \geq 0$ and ν^* such that Equations (6.2) to (6.5) hold. ┘

Proof. The proof is a separating-hyperplane argument applied to first-order feasible directions. Local optimality forbids a feasible first-order descent direction. A constraint qualification lets us replace

the true tangent cone by the linearized cone. Farkas' lemma then converts the absence of a descent direction into nonnegative multipliers on active inequalities and unrestricted multipliers on equalities.

At $\mathbf{x}^*$, local optimality implies that there is no direction $\mathbf{d}$ satisfying

$$\nabla f(\mathbf{x}^*)^\top \mathbf{d} < 0, \quad \nabla g_i(\mathbf{x}^*)^\top \mathbf{d} \leq 0 \text{ for } i \in \mathcal{A}(\mathbf{x}^*), \quad \nabla h_j(\mathbf{x}^*)^\top \mathbf{d} = 0 \text{ for all } j.$$

Under the stated constraint qualification, this absence of linearized feasible descent is valid; otherwise, a linearized descent direction could be integrated or approximated by actual feasible points, contradicting local optimality. By the alternative theorem for systems of linear inequalities, the nonexistence of such $\mathbf{d}$ implies that $-\nabla f(\mathbf{x}^*)$ belongs to the cone generated by active inequality gradients and the span of equality gradients. Thus, there are multipliers $\lambda_i^* \geq 0$ for $i \in \mathcal{A}(\mathbf{x}^*)$ and $\nu_j^* \in \mathbb{R}$ such that

$$\nabla f(\mathbf{x}^*) + \sum_{i \in \mathcal{A}(\mathbf{x}^*)} \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(\mathbf{x}^*) = 0.$$

Set $\lambda_i^* = 0$ for inactive inequalities. Primal feasibility holds because $\mathbf{x}^*$ is feasible. Dual feasibility holds by construction. Complementarity holds because inactive constraints have zero multipliers and active constraints satisfy $g_i(\mathbf{x}^*) = 0$. $\square$

The theorem explains why every algorithm for constrained optimization tries, explicitly or implicitly, to drive the four residuals in [Equations \(6.2\) to \(6.5\)](#) to zero.

7 Convex Sufficiency and Duality

The KKT conditions are necessary under regularity assumptions. In convex optimization they become sufficient. This is the reason KKT is so powerful in convex analysis and machine learning, where many problems are designed so that first-order certificates are global certificates.

Theorem 7.1 (KKT sufficiency for convex programs). Suppose f and g_i are convex, and h_j are affine. If a feasible point $\mathbf{x}^*$ admits multipliers $\lambda^* \geq 0$ and ν^* satisfying the KKT conditions, then $\mathbf{x}^*$ is a global minimizer. $\lrcorner$

Proof. The proof uses convexity to turn stationarity into a global lower bound. The Lagrangian is a global affine-convex underestimator at the stationary point, and feasibility plus complementary slackness make this lower bound equal to the primal objective at $\mathbf{x}^*$.

For any feasible $\mathbf{x}$, convexity of f and g_i gives

$$f(\mathbf{x}) \geq f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*)$$

and

$$g_i(\mathbf{x}) \geq g_i(\mathbf{x}^*) + \nabla g_i(\mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*)$$

for every i . Since h_j is affine, we have

$$h_j(\mathbf{x}) = h_j(\mathbf{x}^*) + \nabla h_j(\mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*).$$

Multiply the inequality for g_i by $\lambda_i^* \geq 0$ and sum the resulting inequalities together with the equality

terms. Using stationarity, we obtain

$$\mathcal{L}(\mathbf{x}, \lambda^*, \nu^*) \geq \mathcal{L}(\mathbf{x}^*, \lambda^*, \nu^*).$$

For feasible $\mathbf{x}$, the inequality constraints satisfy $g_i(\mathbf{x}) \leq 0$ and $\lambda_i^* \geq 0$, while the equalities satisfy $h_j(\mathbf{x}) = 0$. Hence

$$\mathcal{L}(\mathbf{x}, \lambda^*, \nu^*) \leq f(\mathbf{x}).$$

At $\mathbf{x}^*$, complementary slackness and equality feasibility give

$$\mathcal{L}(\mathbf{x}^*, \lambda^*, \nu^*) = f(\mathbf{x}^*).$$

Combining the three displayed relations gives $f(\mathbf{x}) \geq f(\mathbf{x}^*)$ for every feasible $\mathbf{x}$. $\square$

The same proof is the proof of weak duality specialized to a stationary point. The Lagrange dual function

$$g(\lambda, \nu) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \lambda, \nu) \tag{7.1}$$

always gives lower bounds when $\lambda \geq 0$. Under Slater's condition in convex problems, strong duality holds, and KKT characterizes primal-dual optimal pairs [Sla50, Roc70, BV04].

8 Second-Order Conditions

First-order conditions identify stationary candidates. Second-order conditions distinguish minima from maxima and saddle behavior along feasible directions. The relevant directions are not all vectors but the critical cone, which combines active constraints with directions that do not change the objective to first order.

At a KKT point $(\mathbf{x}^*, \lambda^*, \nu^*)$, define the critical cone by

$$\mathcal{C}(\mathbf{x}^*) = \left\{ \mathbf{d} \in \mathcal{L}_{\mathcal{F}}(\mathbf{x}^*) : \nabla f(\mathbf{x}^*)^\top \mathbf{d} = 0 \right\}. \tag{8.1}$$

If the problem is twice continuously differentiable, the Hessian that matters is the Hessian of the Lagrangian

$$\nabla_{\mathbf{xx}}^2 \mathcal{L}(\mathbf{x}^*, \lambda^*, \nu^*). \tag{8.2}$$

Theorem 8.1 (Second-order necessary condition). Let $\mathbf{x}^*$ be a local minimizer satisfying a regularity condition that gives KKT multipliers. Under the standard second-order regularity hypotheses, every critical direction $\mathbf{d} \in \mathcal{C}(\mathbf{x}^*)$ satisfies

$$\mathbf{d}^\top \nabla_{\mathbf{xx}}^2 \mathcal{L}(\mathbf{x}^*, \lambda^*, \nu^*) \mathbf{d} \geq 0. \tag{8.3}$$

Theorem 8.2 (Second-order sufficient condition). Suppose the KKT conditions hold at $\mathbf{x}^*$ and suppose

$$\mathbf{d}^\top \nabla_{\mathbf{xx}}^2 \mathcal{L}(\mathbf{x}^*, \lambda^*, \nu^*) \mathbf{d} > 0 \tag{8.4}$$

for every nonzero critical direction $\mathbf{d} \in \mathcal{C}(\mathbf{x}^*)$. Under the standard constraint qualification hypotheses, $\mathbf{x}^*$ is a strict local minimizer. $\square$

The strict inequality in Equation (8.4) is what gives local curvature in every direction that first-order information cannot already exclude. This is the condition behind local superlinear or quadratic convergence results for Newton and sequential quadratic programming methods [NW06, BGLS06].

9 Complementarity and Sensitivity

Complementary slackness is the part of KKT that has no analogue in equality-constrained Lagrange theory. It says

$$\lambda_i^* g_i(\mathbf{x}^*) = 0 \quad (9.1)$$

for every inequality constraint. Thus, each inequality belongs to one of two regimes. Either the constraint is inactive and its multiplier vanishes, or the constraint is active and the multiplier records the marginal force exerted by the boundary. This dichotomy is the reason active-set methods, complementarity methods, and sensitivity theory all start from the same algebraic identity [CPS92, FP03, NW06].

Definition 9.1 (Strict complementarity). Strict complementarity holds at a KKT point if every active inequality has a positive multiplier, that is,

$$g_i(\mathbf{x}^*) = 0 \implies \lambda_i^* > 0. \quad (9.2)$$

Strict complementarity is not required for KKT necessity or sufficiency, but it often gives the cleanest local picture. If $\mathbf{x}^*$ satisfies LICQ, strict complementarity, and the second-order sufficient condition, then the active set is locally stable under small perturbations in the data, and the constrained problem behaves locally like an equality-constrained problem in which the active inequalities have been correctly guessed [Rob80, BS00, BGLS06]. This is the conceptual reason that an algorithm which eventually identifies the right active set can switch from a combinatorial search to a Newton-like local method.

We can see this stability algebraically. Write the active constraints at $\mathbf{x}^*$ as a vector $\mathbf{c}_{\mathcal{A}}$, collecting the equality constraints and the active inequalities. The local equality-constrained KKT system is

$$\nabla f(\mathbf{x}) + \mathbf{J}_{\mathcal{A}}(\mathbf{x})^\top \mathbf{y} = 0, \quad \mathbf{c}_{\mathcal{A}}(\mathbf{x}) = 0, \quad (9.3)$$

where $\mathbf{J}_{\mathcal{A}}(\mathbf{x})$ is the Jacobian of the active constraints and $\mathbf{y}$ collects the corresponding multipliers. The Jacobian of Equation (9.3) with respect to $(\mathbf{x}, \mathbf{y})$ is the saddle-point matrix

$$\mathbf{K}_{\mathcal{A}} = \begin{pmatrix} \nabla_{\mathbf{xx}}^2 \mathcal{L}(\mathbf{x}^*, \lambda^*, \nu^*) & \mathbf{J}_{\mathcal{A}}(\mathbf{x}^*)^\top \\ \mathbf{J}_{\mathcal{A}}(\mathbf{x}^*) & 0 \end{pmatrix}. \quad (9.4)$$

Under LICQ and the second-order sufficient condition on the critical cone, this matrix is nonsingular on the relevant active manifold. The implicit function theorem then gives differentiable dependence of the primal and dual variables on small perturbations, which is the elementary analytic core behind the more general strong-regularity theory of Robinson [Rob80, BS00].

The multiplier also has an economic and variational meaning. Consider the perturbed problem in which $g_i(\mathbf{x}) \leq u_i$ and $h_j(\mathbf{x}) = v_j$, and let $V(\mathbf{u}, \mathbf{v})$ be the optimal value. Under the usual

differentiability and regularity hypotheses, the envelope relation is

$$\frac{\partial V}{\partial u_i}(0,0) = -\lambda_i^*, \quad \frac{\partial V}{\partial v_j}(0,0) = -\nu_j^*. \quad (9.5)$$

The sign in Equation (9.5) comes from the convention $g_i(\mathbf{x}) \leq u_i$. Relaxing a binding inequality by increasing u_i can only decrease the minimum value, and the multiplier measures that first-order decrease. This is the shadow-price interpretation familiar from linear programming, convex optimization, and parametric nonlinear programming [Roc70, BV04, Ber99, BS00].

10 Algorithms Built Around KKT Systems

Many algorithms for constrained optimization can be read as methods for solving or approximating the KKT system. They differ mainly in how they represent complementarity, how aggressively they enforce feasibility, and how they approximate the Lagrangian Hessian. This point of view is useful because it separates global strategy from local algebra. Line searches, merit functions, filters, and trust regions control global progress, whereas the local step is almost always obtained by solving a linearized or regularized KKT system [Fle87, NW06, BGLS06].

10.1 Active-Set Methods

Active-set methods maintain a working set W_k of constraints that are treated as equalities at the current iterate. Once W_k is fixed, the method computes a step by solving an equality-constrained quadratic model

$$\min_{\mathbf{d}} \quad \nabla f(\mathbf{x}_k)^\top \mathbf{d} + \frac{1}{2} \mathbf{d}^\top \mathbf{H}_k \mathbf{d} \quad \text{subject to} \quad \mathbf{J}_{W_k}(\mathbf{x}_k) \mathbf{d} + \mathbf{c}_{W_k}(\mathbf{x}_k) = 0. \quad (10.1)$$

The equality-constrained KKT equations for Equation (10.1) are

$$\begin{pmatrix} \mathbf{H}_k & \mathbf{J}_{W_k}(\mathbf{x}_k)^\top \\ \mathbf{J}_{W_k}(\mathbf{x}_k) & 0 \end{pmatrix} \begin{pmatrix} \mathbf{d} \\ \mathbf{y} \end{pmatrix} = - \begin{pmatrix} \nabla f(\mathbf{x}_k) \\ \mathbf{c}_{W_k}(\mathbf{x}_k) \end{pmatrix}. \quad (10.2)$$

If the step violates an inactive inequality, the working set is enlarged. If the step reaches a stationary point for the working set but some multiplier has the wrong sign, the corresponding inequality is removed. In quadratic programming this mechanism yields a finite active-set method under nondegeneracy assumptions; in nonlinear programming it becomes a local model inside line-search or trust-region globalization [Fle87, GMW81, NW06, LY16].

10.2 Sequential Quadratic Programming

Sequential quadratic programming, abbreviated SQP, forms at each iterate a quadratic approximation to the Lagrangian and a linear approximation to the constraints. If $\mathbf{H}_k$ approximates $\nabla_{\mathbf{xx}}^2 \mathcal{L}(\mathbf{x}_k, \lambda_k, \nu_k)$, the standard SQP subproblem is

$$\begin{aligned} \min_{\mathbf{d}} \quad & \nabla f(\mathbf{x}_k)^\top \mathbf{d} + \frac{1}{2} \mathbf{d}^\top \mathbf{H}_k \mathbf{d} \\ \text{subject to} \quad & g_i(\mathbf{x}_k) + \nabla g_i(\mathbf{x}_k)^\top \mathbf{d} \leq 0 \quad \text{for } i \in [m], \end{aligned} \quad (10.3)$$

$$h_j(\mathbf{x}_k) + \nabla h_j(\mathbf{x}_k)^\top \mathbf{d} = 0 \quad \text{for } j \in [p].$$

The KKT conditions of Equation (10.3) are precisely the Newton equations for the nonlinear KKT system when the active set is correctly represented and $\mathbf{H}_k$ is the exact Lagrangian Hessian. This explains both the strength and the fragility of SQP. Near a regular solution satisfying second-order sufficiency, exact or well-updated Hessians give fast local convergence; away from such a point, the method needs a merit function, filter, trust region, or penalty mechanism to prevent the quadratic model from pursuing a locally inconsistent linearization [Han77, Pow78, Fle87, NW06, BGLS06].

10.3 Penalty and Augmented Lagrangian Methods

Penalty methods remove constraints by charging for violations, but pure penalty formulations often become ill-conditioned as the penalty parameter grows. Augmented Lagrangian methods keep explicit multiplier estimates and add a moderate quadratic penalty, thereby combining dual first-order information with numerical stabilization [FM68, Hes69, Pow69, Ber99]. For equality constraints, the augmented Lagrangian has the form

$$\mathcal{L}_\rho(\mathbf{x}, \nu) = f(\mathbf{x}) + \nu^\top \mathbf{h}(\mathbf{x}) + \frac{\rho}{2} \|\mathbf{h}(\mathbf{x})\|_2^2. \quad (10.4)$$

After approximately minimizing $\mathcal{L}_\rho(\cdot, \nu_k)$, the multiplier update

$$\nu_{k+1} = \nu_k + \rho \mathbf{h}(\mathbf{x}_{k+1}) \quad (10.5)$$

is a dual ascent step on the penalized model. The method is useful because it does not require feasibility at every inner iteration, while the multiplier estimate still guides the objective gradient toward the normal space of the constraints. Inequality constraints can be handled by slack variables, projected multiplier updates, or specialized Powell–Hestenes–Rockafellar forms [Hes69, Pow69, Ber99, NW06].

10.4 Interior-Point Methods

Interior-point methods replace complementarity by a strictly positive perturbation. For inequalities written as $g_i(\mathbf{x}) + s_i = 0$ with $s_i > 0$, the perturbed primal-dual equations are

$$\nabla f(\mathbf{x}) + \nabla g(\mathbf{x})^\top \lambda + \nabla h(\mathbf{x})^\top \nu = 0, \quad (10.6)$$

$$\mathbf{g}(\mathbf{x}) + \mathbf{s} = 0, \quad (10.7)$$

$$\mathbf{h}(\mathbf{x}) = 0, \quad (10.8)$$

$$\text{Diag}(\mathbf{s})\lambda = \mu \mathbf{1}, \quad \mathbf{s} > 0, \quad \lambda > 0. \quad (10.9)$$

For fixed $\mu > 0$, Equations (10.6) to (10.9) describe a smooth system under regularity assumptions. The central path is the curve of such solutions as μ decreases to zero. Newton’s method applied to this perturbed system gives the primal-dual interior-point step, and the linear algebra again has saddle-point structure, now augmented by diagonal terms coming from $\text{Diag}(\mathbf{s})\lambda = \mu \mathbf{1}$ [Wri97, FGW02, NW06].

In convex optimization, the barrier interpretation gives the same equations from a different

direction. Replacing the inequalities by the logarithmic barrier

$$f(\mathbf{x}) - \mu \sum_{i=1}^m \log(-g_i(\mathbf{x})) \quad (10.10)$$

produces the multiplier relation $\lambda_i = \mu/(-g_i(\mathbf{x}))$, which is equivalent to the perturbed complementarity equation. This identity explains why interior-point methods approach the boundary without touching it, while active-set methods try to identify the boundary explicitly [BV04, Wri97, NW06].

11 Complementarity Problems and Variational Inequalities

KKT conditions can be written as a complementarity problem. If the constraints are $g_i(\mathbf{x}) \leq 0$, introduce slacks $s_i = -g_i(\mathbf{x}) \geq 0$. Complementarity becomes

$$\lambda \geq 0, \quad \mathbf{s} \geq 0, \quad \lambda_i s_i = 0 \text{ for every } i \in [m]. \quad (11.1)$$

Together with stationarity and equality feasibility, Equation (11.1) is a mixed system of equations and complementarity conditions. This is the nonlinear complementarity viewpoint, and it is useful because many equilibrium problems, variational inequalities, contact problems, and market-clearing models have the same algebraic form [CPS92, FP03].

One common way to remove the explicit complementarity relation is to use an NCP function, which is a scalar function $\phi(a, b)$ satisfying

$$\phi(a, b) = 0 \iff a \geq 0, b \geq 0, ab = 0. \quad (11.2)$$

The Fischer–Burmeister function

$$\phi_{\text{FB}}(a, b) = a + b - \sqrt{a^2 + b^2} \quad (11.3)$$

is the standard example [Fis92, FP03]. Applying Equation (11.3) componentwise turns complementarity into nonsmooth equations, which can then be attacked by semismooth Newton methods under suitable regularity assumptions [FP03].

Variational inequalities provide an even broader language. For a closed convex feasible set C , the first-order condition for minimizing differentiable convex f over C is

$$\nabla f(\mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*) \geq 0 \quad \text{for every } \mathbf{x} \in C. \quad (11.4)$$

When C is described by smooth constraints and a constraint qualification holds, Equation (11.4) is equivalent to the KKT conditions. The equivalence comes from the normal-cone inclusion

$$-\nabla f(\mathbf{x}^*) \in \mathcal{N}_C(\mathbf{x}^*), \quad (11.5)$$

where the normal cone of a closed convex set C at $\mathbf{x}^* \in C$ is

$$\mathcal{N}_C(\mathbf{x}^*) = \{\mathbf{y} : \mathbf{y}^\top (\mathbf{x} - \mathbf{x}^*) \leq 0 \text{ for every } \mathbf{x} \in C\}. \quad (11.6)$$

If C is given by differentiable inequalities and equalities, the normal cone in Equation (11.5) is generated by active constraint gradients under the same regularity hypotheses that yield KKT

multipliers. Thus, KKT, complementarity, and variational inequalities are three notations for one geometric fact, namely that the negative objective gradient belongs to the normal cone of the feasible region [Roc70, FP03, BS00].

12 What KKT Teaches

The KKT method teaches that constrained optimization is governed by four simultaneous requirements. A candidate must be feasible for the original problem. Its objective gradient must be balanced by constraint gradients. Inequality multipliers must have the correct sign. A constraint may exert force only when it is tight. These four requirements are simple to state, but the preceding sections show why each of them has a precise geometric role.

In convex optimization, KKT is a global certificate because convexity converts first-order inequalities into global lower bounds. In nonlinear programming, it is a local necessary condition whose validity depends on constraint qualifications. In numerical optimization, it is the nonlinear system that Newton, SQP, augmented Lagrangian, active-set, and interior-point methods try to solve. In sensitivity theory, it explains shadow prices and active-set stability. In complementarity theory, it becomes a structured equation system. The equations are the same, but their force depends on the geometry and regularity of the problem.

The practical lesson is therefore not to memorize the four displayed KKT equations in isolation. We should instead read them as a compact interface between local geometry, convex duality, perturbation theory, and algorithms. Constraint qualifications say when the interface is valid. Second-order conditions say when the stationary point has real local curvature. Complementarity says which inequalities matter. Algorithmic methods differ mainly in how they discover, perturb, or regularize this interface.

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